Module-12a Discrete Fourier Transform

1 Introduction to DFT

In the previous module I have already motivated that most of the signal processing done now a days are on computers. Hence, the signals being processed need to be digital for the computers to understand them. We already know techniques (from Module 11) to transform analog to digital signals. However, in order to do frequency analysis of a discrete-time (digital) signal $x[n]$, we need to take a DTFT of the signal. Recall from module 9a that the DTFT $X(e^{j\omega})$ is continuous and hence, it cannot be handled by a computer. Thus, even though we have learnt to convert a CT signal to a digital signal, we are unable to conduct any frequency analysis of the digital signal until the DTFT is in a digital form. This is the problem that we look into in this module.

In this module we take DT signal $x[n]$ and sample its Fourier transform $X(e^{j\omega})$ to get a discrete-frequency version of $X(e^{j\omega})$. Such a frequency domain representation leads to the Discrete Fourier Transform (DFT). DFT is a powerful computational tool for performing frequency analysis of discrete-time signals.

Frequency Domain Sampling: The DFT

Recall from page 4 module 9a, the analysis equation of DTFT is

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}. \tag{1}$$

We sample $X(e^{j\omega})$ next. Recall that $X(e^{j\omega})$ is periodic with a period of 2π (page 6 module 9a). Hence, we sample only in the fundamental range.

Let us take N equidistant samples of $X(e^{j\omega})$ in the interval $0 \leqslant \omega < 2\pi$. The spacing between the samples is $\frac{2\pi}{N}$ then. Next we decide how to choose N.

Evaluating equation (1) at $\omega = \frac{2\pi}{N}k$,

$$\begin{aligned}
X\left(e^{j\frac{2\pi}{N}k}\right) &= \sum_{n=-\infty}^{\infty} x[n]e^{-j\frac{2\pi}{N}n}, \quad k = 0, 1, \cdots, N-1. \\
&= \cdots + \sum_{n=-N}^{-1} x[n]e^{-j\frac{2\pi}{N}kn} + \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}kn} + \sum_{n=N}^{2N-1} x[n]e^{-j\frac{2\pi}{N}kn} + \cdots \\
&= \sum_{\ell=-\infty}^{\infty} \sum_{n=\ell N}^{\ell N+N-1} x[n]e^{-j\frac{2\pi}{N}kn}.
\end{aligned}$$

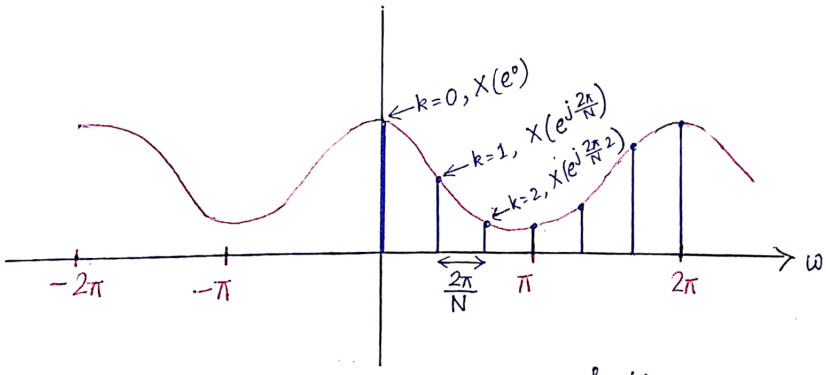


fig (1)

Change the index of sum from n to $n - \ell N$ in the above equation

$$X\left(e^{j\frac{2\pi}{N}k}\right) = \sum_{\ell=-\infty}^{\infty}\sum_{n=0}^{N-1} x[n-\ell N]e^{-j\frac{2\pi}{N}k(n-\ell N)} = \sum_{\ell=-\infty}^{\infty}\sum_{n=0}^{N-1} x[n-\ell N]e^{-j\frac{2\pi}{N}kn} \times e^{j2\pi k\ell}$$

$$= \sum_{\ell=-\infty}^{\infty}\sum_{n=0}^{N-1} x[n-\ell N]e^{-j\frac{2\pi}{N}kn}. \qquad (\text{Since } e^{j2\pi k\ell} = 1)$$

Interchanging the order of summation –

$$X\left(e^{j\frac{2\pi}{N}k}\right) = \sum_{n=0}^{N-1}\underbrace{\left[\sum_{\ell=-\infty}^{\infty} x[n-\ell N]\right]}_{=:x_p[n]} e^{-j\frac{2\pi}{N}kn}, \quad k = 0, 1, \cdots, N-1. \tag{2}$$

Therefore,

$$x_p[n] := \sum_{\ell=-\infty}^{\infty} x[n-\ell N]. \tag{3}$$

The signal $x_p[n]$ is obtained by the periodic repetition of $x[n]$ every N samples. Clearly, $x_p[n]$ is periodic with fundamental period N. Hence, we can apply Fourier series expansion on $x_p[n]$. From module 6a page 10, the Fourier series representation of $x_p[n]$ is –

$$x_p[n] = \sum_{k=0}^{N-1} a_k e^{j\frac{2\pi}{N}kn}, \quad n = 0, 1, \cdots, N-1. \tag{4}$$

$$\text{with } a_k = \frac{1}{N}\sum_{n=0}^{N-1} x_p[n]e^{-j\frac{2\pi}{N}kn}, \quad k = 0, 1, \cdots, N-1. \tag{5}$$

Comparing equation (2) and (5), we have

$$a_k = \frac{1}{N}X\left(e^{j\frac{2\pi}{N}k}\right). \tag{6}$$

Using equation (6) in equation (4), we get

$$x_p[n] = \frac{1}{N}\sum_{k=0}^{N-1} X\left(e^{j\frac{2\pi}{N}k}\right)e^{j\frac{2\pi}{N}kn}, \quad n = 0, 1, \cdots, N-1. \tag{7}$$

Equation (7) gives us a method to reconstruct $x_p[n]$ from $X\left(e^{j\frac{2\pi}{N}k}\right)$ – the frequency domain samples of $X\left(X^{j\omega}\right)$. However this does not help us in getting back what we started with $X\left(e^{j\omega}\right)$ or $x[n]$.

To get back the original signal $x[n]$ we study the relation between $x_p[n]$ and $x[n]$. Remember that $x_p[n]$ is a periodic extension of $x[n]$. Let us look at it pictorially. From Figure (ii), it is

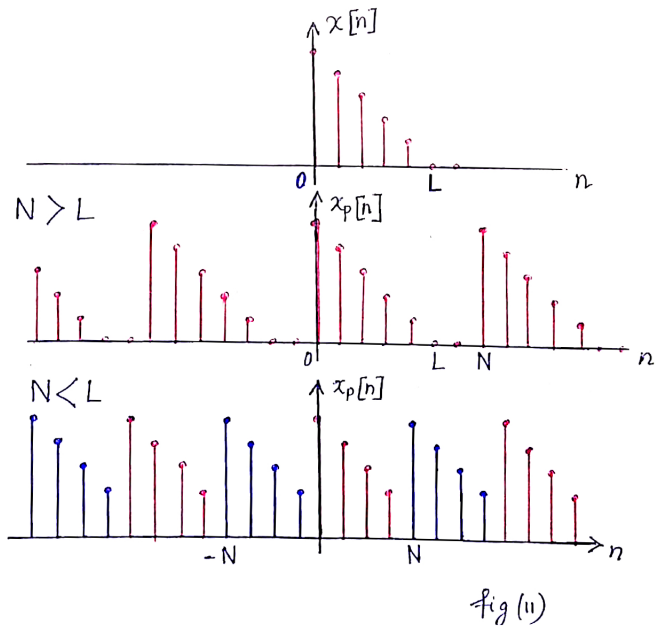


fig (ii)

clear that if $N > L$, then $x[n]$ can be recovered from $x_p[n]$. This means that the original signal has to be time-limited to less than the period N of $x_p[n]$.

This is analogous to the time-sampling case. Recall that in time-sampling the original signal has to be band-limited and the maximum frequency must be less than half of the sampling frequency for exact reconstruction.

Time-sampling	$\omega_m < \dfrac{\omega_s}{2}$ (Condition is on frequency).
Frequency-sampling	$L < N$ (Condition is on time).

$N < L$ leads to time-domain aliasing – clear from Figure (ii). Thus,

The spectrum of an aperiodic discrete-time signal with finite duration L can be exactly recovered from its samples at frequencies $\omega_k := \frac{2\pi k}{N}$, $k = 0, 1, \cdots, N-1$, if $N > L$. The procedure to compute $x[n]$ from $x_p[n]$ is

$$x[n] = \begin{cases} x_p[n], & \text{for } 0 \leqslant n \leqslant N-1 \\ 0, & \text{for otherwise.} \end{cases} \tag{8}$$

Then we equation (8) to get $X(e^{j\omega})$.

Thus, we need to look at finite sequences of length $L \leqslant N$ only if we are to reconstruct the signal successfully from the samples of the frequency spectrum.

Therefore, equation (1) can be rewritten as –

$$X(e^{j\omega}) = \sum_{n=0}^{L-1} x[n]e^{-j\omega n} \tag{9}$$

On sampling $X(e^{j\omega})$ at equally spaced frequencies $\omega_k = \frac{2\pi}{N}k$, $k = 0, 1, \cdots, N-1$, where $N \geqslant L$, we get

$$X[k] := X(e^{j\omega_k}) = \sum_{n=0}^{L-1} x[n]e^{-j\omega_k n}$$

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j\omega_k n} \tag{10}$$

We have changed the upper limit of the sum in equation (10) to $N-1$ from $L-1$. This is done for convenience. Note that this change does not effect equation (9) because $x[n] = 0$ for $n \geqslant L$.

Since the frequency sample are obtained by Fourier transform $X(e^{j\omega})$ at a set of N discrete frequencies, equation (10) is called discrete Fourier transform (DFT) of $x[n]$. From equation (8) we infer that $x[n]$ can be recovered from $X[k]$ by the formula:

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]e^{j\frac{2\pi}{N}kn}. \tag{11}$$

To summarize

DFT (Discrete Fourier Transform)

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}kn}, \quad k = 0, 1, \cdots, N-1. \tag{12}$$

IDFT (Inverse Discrete Fourier Transform)

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]e^{j\frac{2\pi}{N}kn}, \quad n = 0, 1, \cdots, N-1. \tag{13}$$

Note that DFT and DTFT is different. In DTFT we take any length (finite/infinite) of signal and the spectrum is continuous. DFT is derived from DTFT by sampling $X(e^{j\omega})$.

DTFT: $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$ DFT: $X[k] = \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}kn}, \quad k = 0, 1, \cdots, N-1.$

Equation (12) is also known as the N-point DFT of $x[n]$ – meaning $X(e^{j\omega})$ evaluated at the set of N equally spaced frequencies $\omega_k = \dfrac{2\pi}{N}$, where $k = 0, 1, \cdots, N-1$.

Example 1.1 *Determine the N-point DFT of the sequence*

$$x[n] = \begin{cases} 1, & 0 \leqslant n \leqslant L-1 \\ 0, & n \geqslant L. \end{cases}$$

for $N \geqslant L$.

Solution: *From equation* (12),

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}kn} = \sum_{n=0}^{L-1} e^{-j\frac{2\pi}{N}kn} = \frac{1-e^{-j\frac{2\pi}{N}Lk}}{1-e^{-j\frac{2\pi}{N}k}} = \frac{\sin(\frac{\pi kL}{N})}{\sin(\frac{\pi k}{N})} \quad k = 0, 1, \cdots, N-1.$$

Now if we choose $N = L$, then

$$X[k] = \begin{cases} L, & k = 0 \\ 0, & k = 1, 2, \cdots, L-1. \end{cases}$$

Thus there is only one nonzero value in the DFT.

Can we get back the original signal $x[n]$ with just one value of the spectrum? Let us use the IDFT formula to check.

$$x[n] = \frac{1}{N}\sum_{k=0} N-1X[k]e^{j\frac{2\pi}{N}kn} \quad \textit{for } n = 0, 1, \cdots, N-1$$

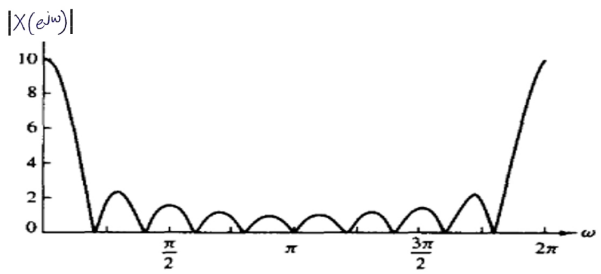
$$\Rightarrow x[n] = \frac{1}{N}X[0] = \frac{L}{N} = \frac{N}{N} = 1 \quad \textit{for } n = 0, 1, \cdots, N-1.$$

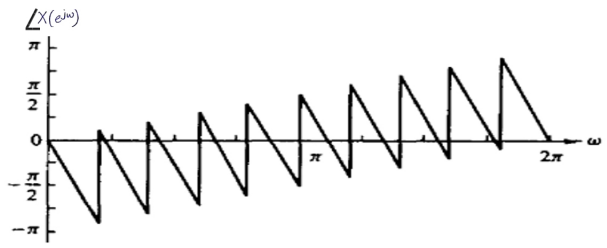
Thus, we have the original signal back.

For $N \neq L$, if we had done a DTFT of $x[n]$ then

$$X(e^{j\omega}) = \frac{\sin(\frac{\omega L}{2})}{\sin(\frac{\omega}{2})}e^{-j\omega(L-1)/2}.$$

The spectrum would look as follows:



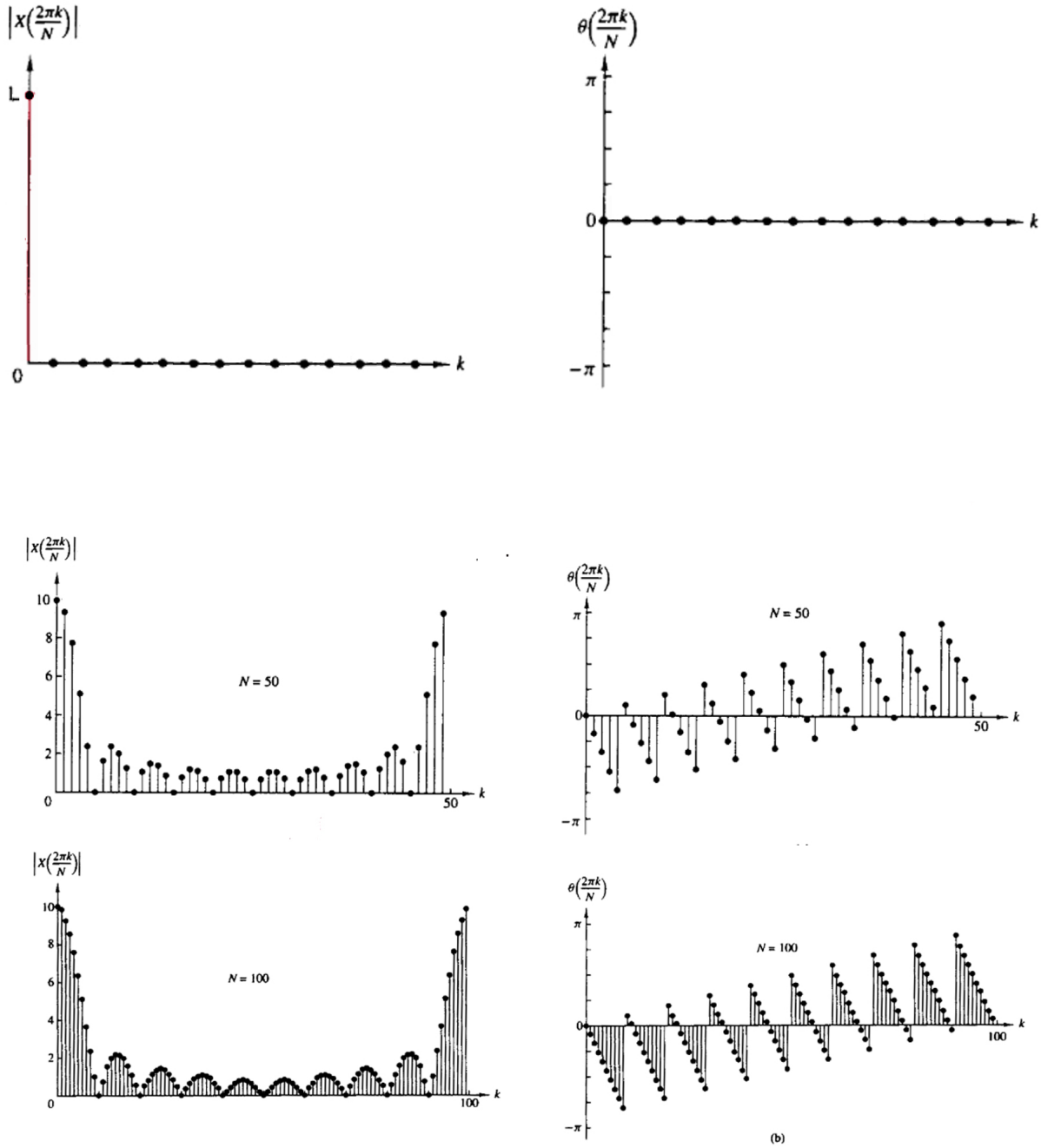


At $N = L$ we have the following spectrum:

Clearly, at $N = L$ we do not get sufficient details of the spectrum when compared to $X(e^{j\omega})$. However, if we sample at closer frequencies we will get much better details of the spectrum. This, of course, comes at the cost of handling more data.

The DFT can easily be written down in matrix representation. This helps in coding it easily into a computer.

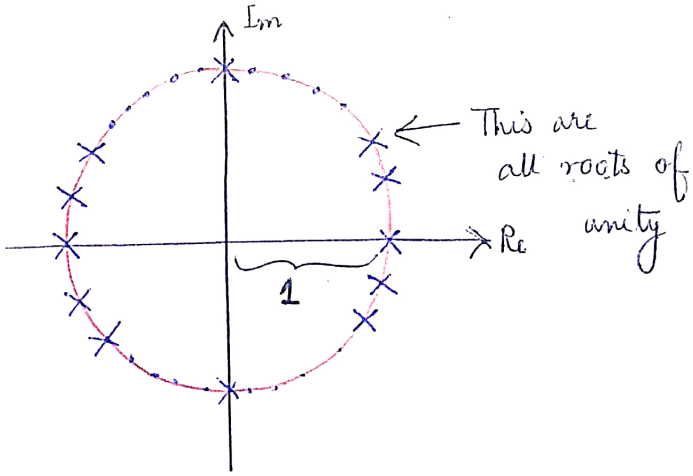
Define $W_N := e^{-j\frac{2\pi}{N}}$. Rewriting equations (12) and (13) in terms of W_N, we get



$$\text{DFT:} \quad X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}, \quad k = 0, 1, \cdots, N-1. \tag{14}$$

$$\text{IDFT:} \quad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}, \quad n = 0, 1, \cdots, N-1. \tag{15}$$

Significance of W_N: W_N is the N-th root of unity, i.e., if we solve the equation $r^N = 1$, then there will be N roots, and W_N is the N-th root of unity.



Now consider the N-point vectors $\mathtt{x}_{\mathtt{N}}$ and X_{N} defined as –

$$\mathtt{x}_{\mathtt{N}} := \begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[L-1] \\ x[L] \\ \vdots \\ x[N-1] \end{bmatrix}$$

$\mathtt{x}_{\mathtt{N}}$ is formed by the signal values at different instants. If $L < N$, then $x[L], x[L+1], \cdots, x[N-1]$ will be zero. $L > N$ is not considered because this will lead to aliasing.

$$\mathrm{X}_{\mathrm{N}} := \begin{bmatrix} X[0] \\ X[1] \\ \vdots \\ X[N-1] \end{bmatrix}$$

Vector X_{N} comprises of the sample values of the DTFT $X(e^{j\omega})$ sampled equidistantly at a distance of N.

We define the following $N \times N$ matrix

$$\mathbb{W}_N := \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & W_N & W_N^2 & \cdots & W_N^{N-1} \\ 1 & W_N^2 & W_N^4 & \cdots & W_N^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & W_N^{N-1} & W_N^{2(N-1)} & \cdots & W_N^{(N-1)(N-1)} \end{bmatrix} \tag{16}$$

Then equation (14) written in the matrix form gives

$$\mathrm{X}_{\mathrm{N}} = \mathbb{W}_N \mathtt{x}_{\mathtt{N}}. \tag{17}$$

If $\mathbb{W}_N$ is assumed to be invertible (which is true), then

$$\mathtt{x_N} = \mathbb{W}_N^{-1}\mathrm{X_N}. \tag{18}$$

Thus, equation (18) gives us a method to compute the original signal $x[n]$ from the frequency spectrum samples $X[k]$. This is what equation (15) does. Therefore, equation (18) is the IDFT equation in matrix form. You must be wondering where did the $\frac{1}{N}$ term in equation (15) go then?

Let us start from equation (15) then. If equation (15) is written in matrix form we have

$$\mathtt{x_N} = \frac{1}{N}\begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & W_N^{-1} & W_N^{-2} & \cdots & W_N^{-(N-1)} \\ 1 & W_N^{-2} & W_N^{-4} & \cdots & W_N^{-2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & W_N^{-(N-1)} & W_N^{-2(N-1)} & \cdots & W_N^{-(N-1)(N-1)} \end{bmatrix}\begin{bmatrix} X[0] \\ X[1] \\ X[2] \\ \vdots \\ x[N-1] \end{bmatrix}$$

$$\Rightarrow \mathtt{x_N} = \frac{1}{N}\begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & W_N^{*} & (W_N^{2})^{*} & \cdots & (W_N^{(N-1)})^{*} \\ 1 & (W_N^{2})^{*} & (W_N^{4})^{*} & \cdots & (W_N^{2(N-1)})^{*} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & (W_N^{(N-1)})^{*} & (W_N^{2(N-1)})^{*} & \cdots & (W_N^{(N-1)(N-1)})^{*} \end{bmatrix}\begin{bmatrix} X[0] \\ X[1] \\ X[2] \\ \vdots \\ x[N-1] \end{bmatrix}$$

$$\Rightarrow \mathtt{x_N} = \frac{1}{N}\begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & W_N^{-1} & W_N^{-2} & \cdots & W_N^{-(N-1)} \\ 1 & W_N^{-2} & W_N^{-4} & \cdots & W_N^{-2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & W_N^{-(N-1)} & W_N^{-2(N-1)} & \cdots & W_N^{-(N-1)(N-1)} \end{bmatrix}^{*}\begin{bmatrix} X[0] \\ X[1] \\ X[2] \\ \vdots \\ x[N-1] \end{bmatrix}$$

Therefore,

$$\mathtt{x_N} = \frac{1}{N}\mathbb{W}_N^{*}\mathrm{X_N} \tag{19}$$

Comparing equation (18) and (19), we have

$$\mathbb{W}_N^{-1} = \frac{1}{N}\mathbb{W}_N^{*}. \tag{20}$$

We have not explicitly proved that the inverse of $\mathbb{W}_N$ exists. But notice that equation (19) is always true, since it is derived from the IDFT equations (13) and (15). Hence, $\mathtt{x_N}$ can be obtained from $\mathrm{X_N}$ by equation (19) and therefore $\mathbb{W}_N^{-1}$ must exist. $\mathbb{W}_N^{-1}$ is thus related to $\mathbb{W}_N$ by equation (20). There are other ways of proving that $\mathbb{W}_N$ is invertible. However those methods, based on basic linear algebra, are not in the scope of this course. In summary,

$$\text{DFT: } \mathrm{X_N} = \mathbb{W}_N \mathtt{x_N} \qquad \text{IDFT: } \mathtt{x_N} = \mathbb{W}_N^{-1}\mathrm{X_N} = \frac{1}{N}\mathbb{W}_N^{*}\mathbb{W}_N\mathrm{X_N},$$

where $\mathbb{W}_N$ is as defined in equation (16).

Let us consider the previous example for $L = 2$ now and say we want to find 3-point DFT.

Example 1.2 *The sequence is* $x[n] = \{1, 1, 0\}$ *and* $\mathbb{W}_3 = e^{-j\frac{2\pi}{3}}$.

$$\begin{bmatrix} X[0] \\ X[1] \\ X[2] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & W_3 & W_3^2 \\ 1 & W_3^2 & W_3^4 \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & e^{-j\frac{2\pi}{3}} & e^{-j\frac{4\pi}{3}} \\ 1 & e^{-j\frac{4\pi}{3}} & e^{-j\frac{8\pi}{3}} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 + e^{-j\frac{2\pi}{3}} \\ 1 + e^{-j\frac{4\pi}{3}} \end{bmatrix}.$$

What if $N = 2$. *Then* $W_2 = e^{-j\frac{2\pi}{2}} = -1$.

$$\begin{bmatrix} X[0] \\ X[1] \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & W_2 \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

This is exactly what we expect as seen in the previous example.

Note, given a sequence we can have different point DFTs. The point denotes the number of samples of $X(e^{j\omega})$ taken between 0 and 2π.

From the analysis till this point, it is clear that DFT is an important computational tool for performing frequency analysis of signals on digital signal processors. Once we have the finite sequence we can very easily feed it into the computer and find the DFT using simple matrix operations. Till now we have read about many other tools like CTFT, DTFT, and z-Transform. If we manage to establish a way to compute those using DFT, then writing codes to find them might be easier. Hence, we try to develop these relations next.

- **Relation between DFT and DTFS:** A periodic sequence $x_p[n]$ with fundamental period N has the following DTFS (Module 6a) –

$$x_p[n] = \sum_{k=0}^{N-1} a_k e^{j\frac{2\pi}{N}kn} \qquad \text{(Equation (A), Module 9b)} \tag{21}$$

$$\text{where } a_k = \frac{1}{N}\sum_{n=0}^{N-1} x_p[n] e^{-j\frac{2\pi}{N}kn} \text{ (Equation (B), Module 9b).} \tag{22}$$

  Note that equation (21) and equation (13) are similar. Equation (22) and equation (12) are similar as well.

  From the comparison it is clear that if we define a sequence $x[n]$ from the periodic sequence $x_p[n]$, such that

$$x[n] = \begin{cases} x_p[n], & 0 \leqslant n \leqslant N-1, \\ 0, & \text{otherwise,} \end{cases} \tag{23}$$

then the DFT of $x[n]$ is Na_k. In other words, $X[k] = Na_k$, $k = 0, 1, \cdots, N-1$.

Note that if we put $a_k = \dfrac{X[k]}{N}$ in equation (21), we get

$$x_p[n] = \sum_{k=0}^{N-1} \frac{1}{N} X[k] e^{j\frac{2\pi}{N}kn} = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi}{N}kn}.$$

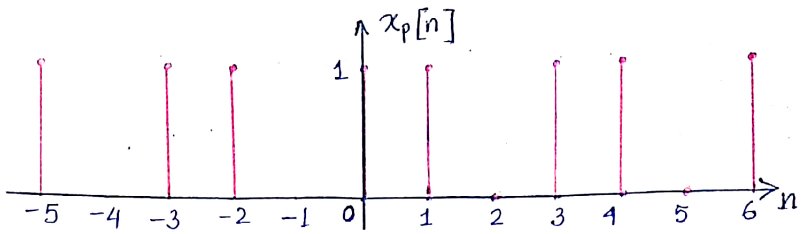
This is the IDFT equation. This means if we take DFT of $x[n]$ (one period of $x_p[n]$) and multiply the samples by $\frac{1}{N}$, we get the Fourier series coefficients of the periodic sequence $x_p[n]$. And just taking a IDFT of the sequence $X[k]$ gives the periodic sequence $x_p[n]$.

Should not the IDFT of $X[k]$ sequence give back $x[n]$?

Yes, on using equation (13), we do get back $x[n]$. However, if we repeat $x[n]$ with a period of N , then we get $x_p[n]$, i.e., $x_p[n] = \sum_{\ell=-\infty}^{\infty} x[n - \ell N]$.

Let us consider a simple example to illustrate what we discussed.

Example 1.3 *Consider the following periodic sequence with period* 2.



We first find the Fourier series coefficients.

$$a_k = \frac{1}{N} \sum_{n=0}^{N-1} x_p[n] e^{-j\frac{2\pi}{N}n}.$$

Here $N = 3$ *and* $x_p[0] = x_p[1] = 1$, $x_p[2] = 0$. *Therefore,*

$$a_k = \frac{1}{3} \sum_{n=0}^{2} x_p[n] e^{-j\frac{2\pi}{N}kn}.$$

Thus, the spectral coefficients are:

$$\begin{aligned}
a_0 &= \frac{1}{3}\left(x_p[0] + x_p[1]\right) = \frac{2}{3}. \\
a_1 &= \frac{1}{3}\left(x_p[0] + x_p[1]e^{-j\frac{2\pi}{3}}\right) = \frac{1}{3}\left[1 + e^{-j\frac{2\pi}{3}}\right] \\
a_2 &= \frac{1}{3}\left(x_p[0] + x_p[1]e^{-j\frac{4\pi}{3}}\right) = \frac{1}{3}\left[1 + e^{-j\frac{4\pi}{3}}\right].
\end{aligned}$$

So, we have the FS coefficients. Using these coefficients we can again reconstruct back $x_p[n]$ *by using equation* (2) *in Module 6a.*

Now we use DFT to find the Fourier series coefficients. To use DFT we form a sequence just taking one period's data $x_p[n]$. We define

$$x[n] = \begin{cases} x_p[n], & 0 \leqslant n \leqslant N-1, \\ 0, & \textit{otherwise.} \end{cases}$$

Now we find N-point DFT, i.e., 3-point DFT of $x[n]$. Here $W_3 = e^{-j\frac{2\pi}{3}}$.

$$\begin{bmatrix} X[0] \\ X[1] \\ X[2] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & e^{-j\frac{2\pi}{3}} & e^{-j\frac{4\pi}{3}} \\ 1 & e^{-j\frac{4\pi}{3}} & e^{-j\frac{8\pi}{3}} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1+e^{-j\frac{2\pi}{3}} \\ 1+e^{-j\frac{4\pi}{3}} \end{bmatrix}.$$

Therefore, $a_k = \dfrac{X[k]}{N}$, i.e.,

$$\begin{bmatrix} X[0] \\ X[1] \\ X[2] \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 \\ 1+e^{-j\frac{2\pi}{3}} \\ 1+e^{-j\frac{4\pi}{3}} \end{bmatrix}.$$

Thus, DFT can be used to compute the spectrum of a periodic sequence.

- **Relation between DTFT and DFT:** We have already seen that if $x[n]$ is an aperiodic finite energy sequence with FT $X(e^{j\omega})$, which is sampled at N equally spaced frequencies $\omega_k = \dfrac{2\pi k}{N}$, $k = 0, 1, \cdots, N-1$, the spectral components

$$X[k] = X(e^{j\omega})|_{\omega=\frac{2\pi}{N}k} = \sum_{n=-\infty}^{\infty} x[n] e^{1j\frac{2\pi}{N}kn}, \quad k = 0, 1, \cdots, N-1$$

are the DFT coefficients of the periodic sequence of period N, given by

$$x_p[n] = \sum_{\ell=-\infty}^{\infty} x[n - \ell N].$$

Thus, $x_p[n]$ is determined by aliasing $x[n]$ over the interval $0 \leqslant n \leqslant N-1$. The finite duration sequence

$$\hat{x}[n] = \begin{cases} x_p[n], & 0 \leqslant n \leqslant N-1 \\ 0, & \text{otherwise,} \end{cases}$$

bears no resemblance to the original sequence $x[n]$, unless $x[n]$ is of finite duration and length $L \leqslant N$, in which case:

$$x[n] = \hat{x}[n], \qquad 0 \leqslant n \leqslant N-1.$$

Only in this case will IDFT of $X[k[$ yield $x[n]$.

Although we have established a link between DTFT and DFT, nowhere have we mentioned if it is possible to get $X(e^{j\omega})$ at arbitrary frequencies from $X[k]$ – may be we can or may be we cannot – let us move forward and see.

- **Relation between DFT and z-Transform:** If $x[n]$ is a finite-duration signal of length L say and $N \geqslant L$, then the z-Transform of $x[n]$ can be written as:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} = \sum_{n=0}^{L-1} x[n]z^{-n} = \sum_{n=0}^{N-1} x[n]z^{-n}.$$

Using the IDFT formula of $x[n]$ from equation (13) in the above equation, we get

$$\begin{aligned}
X(z) &= \sum_{n=0}^{N-1}\left[\frac{1}{N}\sum_{k=0}^{N-1} X[k]e^{j\frac{2\pi}{N}kn}\right] z^{-n} \\
&= \frac{1}{N}\sum_{n=0}^{N-1}\sum_{k=0}^{N-1} X[k]e^{j\frac{2\pi}{N}kn}z^{-n} = \frac{1}{N}\sum_{k=0}^{N-1} X[k]\sum_{n=0}^{N-1}\left(e^{j\frac{2\pi}{N}k}z^{-1}\right)^n \\
\Rightarrow X(z) &= \frac{1}{N}\sum_{k=0}^{N-1} X[k]\left[\frac{1-(e^{j\frac{2\pi}{N}k}z^{-1})^N}{1-e^{j\frac{2\pi}{N}k}z^{-1}}\right] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]\left[\frac{1-z^{-N}}{1-e^{j\frac{2\pi}{N}k}z^{-1}}\right]
\end{aligned}$$

Therefore,

$$X[k] = \left(\frac{1-z^{-N}}{N}\right)\sum_{k=0}^{N-1}\frac{X[k]}{1-e^{j\frac{2\pi}{N}k}z^{-1}}. \tag{24}$$

When evaluated on the unit circle, equation (24) yields Fourier transform of $x[n]$ in terms of DFT, i.e., at $z = e^{j\omega}$

$$X(e^{j\omega}) = \left(\frac{1-e^{-j\omega N}}{N}\right)\sum_{k=0}^{N-1}\frac{X[k]}{1-e^{-j(\omega-\frac{2\pi}{N}k)}} \tag{25}$$

Equation (25) is a polynomial interpolation formula for $X(e^{j\omega})$ in terms of $X[k]$.

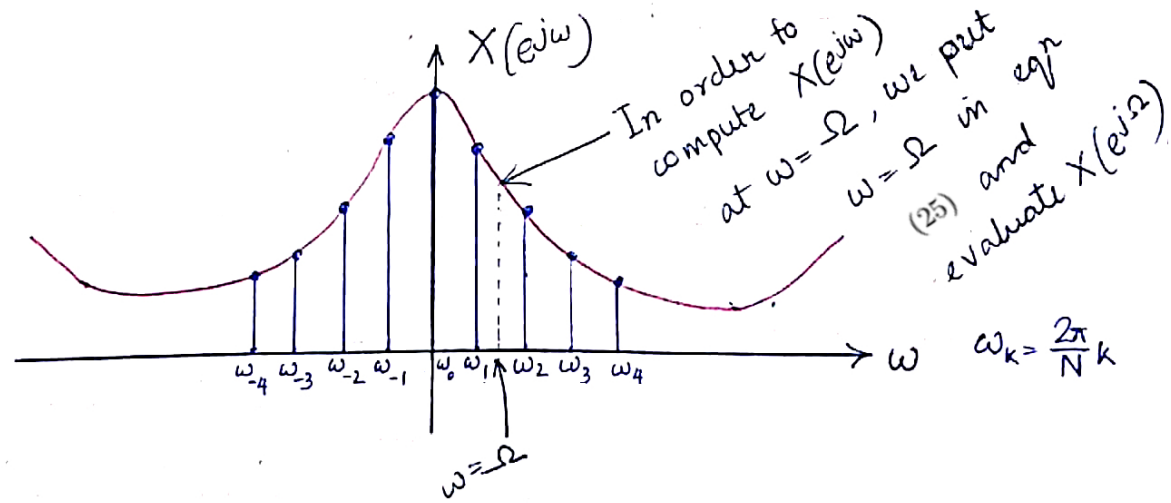
Interpolation formulas are used to find unknown values of a function using the values of the function known at a few points.

A simple example is that of linear interpolation. Suppose that two points in the (x, y) co-ordinate are given to you – (x_1, y_1) and (x_2, y_2). Further you have the information that the points lie on a straight line. Then then interpolation formula is –

$$\frac{y-y_1}{x-x_1} = \frac{y_2-y_1}{x_2-x_1} \tag{26}$$

Using equation (26), you have a formula to compute all the points in the line joining (x_1, y_1) and (x_2, y_2).

Note that DFT are the sampled points of $X(e^{j\omega})$ and in order to find the value of $X(e^{j\omega})$ at any point between the sampled values, we can use equation (25). Thus, we can compute



DTFT at any arbitrary frequency using DFT and equation (25).

- **Relation between CTFS and DFT:** Suppose that $x_a(t)$ is a continuous-time periodic signal with fundamental period $T_p = \frac{1}{F_0}$.

$$x_a(t) = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi}{T_p}kt} = \sum_{k=-\infty}^{\infty} a_k e^{j2\pi F_0 kt},$$

where a_k are the Fourier coefficients. If we sample $x_a(t)$ at a uniform rate $F_s = \frac{N}{T_p} = \frac{1}{T}$, we get

$$x[n] := x(nT) = \sum_{k=-\infty}^{\infty} a_k e^{j2\pi F_0 knT} = \sum_{k=-\infty}^{\infty} a_k e^{j2\pi F_0 kn\frac{1}{NF_0}} = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi}{N}kn}$$

$$\Rightarrow x[n] = \sum_{k=0}^{N-1}\left[\sum_{\ell=-\infty}^{\infty} a_{k-\ell N}\right] e^{j\frac{2\pi}{N}kn}. \tag{27}$$

Clearly, equation (27) is similar to the IDFT equation (13). Therefore, the relation between CTFS coefficients and DFT is

$$X[k] = N\sum_{\ell=-\infty}^{\infty} a_{k-\ell N}.$$

Thus, in this module, we learned how to compute DFT and use it to compute frequency spectrum of a finite-duration discrete-time sequence. In the next module we look at the properties of DFT.